B.Tech / M.Tech - Project / Dissertation Proposal Submission - AY 2026 - 2027

*This form collects the project / dissertation proposal for screening by the committee. * For B.Tech batches, it is mandatory to fill all student names and roll nos. * For M.Tech, please ignore student 2 & 3. *The submission form will be open till **30 April 2026, 1700 Hrs**.

1. Batch ID *

ECE039

2. Student 1 - Name *

Laxmi Bhargavi U

3. Student 1 - Roll No. *

CB.EN.U4CCE23060

4. Student 2 - Name

Hrudyasree ch

5. Student 2 - Roll No.

CB.EN.U4CCE23007

6. Student 3 - Name

Jaiwanthi R

7. Student 3 - Roll No.

CB.EN.U4CCE23017

8. Faculty Mentor *

Anusha K S

9. Title of Project Proposal *

Resilient Multi-Hop Wireless Communication System with Intelligent Emergency Response

10. Objectives [in Bullet form] *

To design a resilient, self-healing multi-hop wireless network capable of dynamically routing around failed nodes and prioritizing critical emergency signals (e.g., SOS alerts) to ensure guaranteed, low-latency transmission.

To implement lightweight, on-node data processing to locally filter environmental sensor data, thereby minimizing network bandwidth congestion and optimizing power consumption during non-emergencies.

To develop an offline, intelligent gateway system that utilizes machine learning to continuously analyze incoming multi-sensor data, detecting both known hazards and unexpected environmental anomalies without reliance on

11. Keyword *

[Choose the keyword that best describes your project work. ML/AI is a tool that can be sued in any of the disciplines mentioned below]

- ○ AD / ADAS
- ○ Circuits and Instrumentation
- ○ Battery Management Systems
- ○ Biomedical Systems
- ○ Electric vehicles and Electrification
- ○ Embedded Systems & IoT
- ○ Radio Frequency Engineering
- ○ Signal, Image and Video Processing
- ○ Software Defined Vehicles
- ○ VLSI Design, testing and Security
- ◉ Wireless Communication Networks and Security

12. Milestones and Deliverables: June - July 2026 *

Milestone: Requirement Analysis & Hardware Setup

Deliverables: Finalization of the system architecture and block diagrams. Procurement of microcontrollers, wireless transceivers, and environmental

13. Milestones and Deliverables: August - September 2026 *

Milestone: Multi-Hop Network Development & Routing Logic

Deliverables: Implementation of the wireless communication protocol between nodes. Development of the priority routing logic for emergency signals and the self-healing algorithm to automatically bypass failed or

14. Milestones and Deliverables: October - November 2026 *

Milestone: On-Node Processing & Machine Learning Training

Deliverables: Deployment of lightweight data filtering algorithms directly on the sensor nodes to reduce bandwidth usage. Preparation of the environmental dataset. Training of the core machine learning models on the

15. Milestones and Deliverables: December 2026 - January 2027 *

Milestone: Transparent Decision Layer & Dashboard Integration

Deliverables: Integration of the transparent reasoning logic to generate clear, human-readable alerts (e.g., explaining why an alarm is triggered).

16. Milestones and Deliverables: February 2027 - March 2027 *

Milestone: System Testing, Validation & Final Documentation

Deliverables: Comprehensive stress testing of the fully integrated system by simulating physical node failures and emergency disaster conditions. Evaluation of key metrics (network latency, packet delivery ratio, and AI

17. Targeted Conference / Journal for Publication *

Targeted IEEE Conferences (Scopus Indexed):
IEEE WiSPNET: International Conference on Wireless Communications, Signal Processing and Networking.
IEEE ICCSP: International Conference on Communications and Signal Processing.
IEEE ANTS: International Conference on Advanced Networks and Telecommunications Systems.

18. Tools used *

Hardware and Software – Name, Specs and Availability in Campus

Hardware Requirements:

Microcontrollers: ESP32 Development Boards.
Sensors & Inputs: DHT11 modules, MQ2 sensors, and basic push-buttons.
Central Gateway Server: laptop.

Software Requirements:

Node Programming: Arduino IDE (using C/C++).
Machine Learning Development: Python 3, utilizing standard libraries like Scikit-learn and TensorFlow/Keras.
Hardware-to-PC Communication: The PySerial Python library.

19. Abstract of the proposal *

Traditional disaster networks rely on cloud infrastructure, which fails during severe damage. This project proposes an entirely offline, resilient multi-hop Wireless Sensor Network (WSN) featuring an intelligent, closed-loop decision layer. To prevent bandwidth congestion, hardware nodes perform lightweight on-device filtering, transmitting only critical updates. The system utilizes an AI-driven adaptive sampling mechanism, where the gateway actively adjusts the sleep cycles of remote nodes based on real-time risk, optimizing battery

20. Proposed Novelty / Contribution *

This idea is fully offline, closed-loop WSN architecture solving current limitations via three pillars:
• Distributed Processing: On-node filtering using AI saves bandwidth, while the gateway uses Explainable AI (XAI) for transparent alerts.
• Context-Aware Networking: The system dynamically adapts node sleep

21. Details of Datasets (if any) *

Name, Dimension, Availability, Purpose of use, Authenticity

Name: IoT Environmental and Smoke Detection Dataset
Dimension: Approx. 62,600 rows (records) and 15 columns (features including Temperature, Humidity, eCO2, TVOC gas levels, and Fire triggers).
Availability: Open-source and freely available to download from Kaggle / Mendeley Data.
Purpose of use: To train the central gateway to recognize the baseline for "normal" environmental conditions versus hazardous anomalies. This data is required for the system to accurately trigger automated XAI alerts and to

22. References *

Most relevant five References from journals with Scopus Percentile >80 in IEEE Format - in bulleted form.

--X. Liu, Y. Zhang, and J. Wang, "A Resilient Multi-Hop Routing Strategy for Wireless Sensor Networks in Underground Environments," IEEE Sensors Journal, vol. 22, no. 14, pp. 14500-14512, Jul. 2022.

--M. A. Rahman, M. S. Hossain, and G. Muhammad, "Artificial Intelligence-Based Real-Time Disaster Management Framework Using Edge-IoT," IEEE Internet of Things Journal, vol. 9, no. 11, pp. 8321-8330, Jun. 2022.

--S. Bera, S. Misra, and N. S. Raghuwanshi, "AI-Envisioned Edge-Assisted Resilient Communication in Post-Disaster Environments," IEEE Transactions on Network and Service Management, vol. 20, no. 1, pp. 456-468, Mar. 2023.

--H. Yang, X. Li, and C. Jiang, "Machine Learning-Empowered Emergency Data Forwarding in Multi-Hop IoT Networks," IEEE Transactions on Industrial Informatics, vol. 19, no. 2, pp. 1250-1260, Feb. 2023.

23. Declaration *

I declare that the proposal submitted is complete in all aspects and is based on the discussion with my faculty mentor and team members. I declare that the proposal is not AI generated / copied from internet sources. I confirm that I shall provide a presentation of the proposal with all the necessary details if required by the review committee. I understand that this proposal will be considered for review only on submitting the hardcopy of the proposal duly endorsed by all the team members and faculty mentor by the due date.

- [x] Agree

24. Please sign here after printing

Student 1 Bhargavi U

Student 2

Student 3

Faculty Mentor